

Che CHENG

MPhil Student, Computer and Information Engineering
The Chinese University of Hong Kong, Shenzhen
Shenzhen, China | checheng@link.cuhk.edu.cn | [Homepage](#) | [GitHub](#)

PROFILE

PhD-seeking MPhil student in embodied AI, robotic perception, and multimodal learning. My work focuses on structured 3D grounding, temporal interaction understanding, and geometry-aware multimodal perception. I am interested in building reliable perception-to-decision interfaces that expose intermediate state, uncertainty, and diagnostic signals for embodied systems operating under occlusion, changing scene state, and limited supervision.

EDUCATION

The Chinese University of Hong Kong, Shenzhen 2025.09–2027.06
MPhil in Computer and Information Engineering
Research at iHEAR Lab on 3D vision, surgical automation, embodied intelligence, multimodal learning, and robotic perception/control.
Advisor: [Prof. Fangxun Zhong](#).

Dalian Maritime University 2021.09–2025.06
Bachelor of Management in Big Data Management and Application
Minor in Software Engineering leading to a Bachelor of Engineering degree.

RESEARCH AGENDA

- **Structured 3D Grounding:** grounding objects, regions, spatial relations, and task context into structured representations for embodied perception.
- **Temporal Interaction Understanding:** modeling how agent-scene relationships evolve in dynamic, partially observable, and visually ambiguous scenes.
- **Reliable Embodied Decision Interfaces:** designing intermediate representations and evaluation protocols that make perception-driven decisions easier to inspect, diagnose, and improve.

SELECTED OPEN-SOURCE PROJECTS

Spatial Prism: Relation-Aware 3D Grounding for Embodied Perception [GitHub](#)

- Built a reproducible diagnostic project for relation-aware 3D visual grounding with scene-disjoint Nr3D/ReferIt3D-style evaluation, hard-subset analysis, coverage diagnostics, and controlled relation-scoring ablations.
- Preserved claim boundaries in the public report: reproduced baseline Acc@1/Acc@5, analyzed same-class clutter and high-clutter subsets, and reported controlled ablations without making state-of-the-art claims.
- Public artifacts include a course report, reproducibility notes, evidence map, diagnostic summaries, coverage reports, and aggregate tables; the README reports a 4,255-sample scene-disjoint baseline and protocol-specific hard-subset diagnostics.
- The public summary separates reproduced baselines, frozen-logit diagnostics, trained controlled ablations, pilots, and negative findings to keep evidence levels auditable.

Keywords: 3D grounding; relation reasoning; embodied perception; diagnostics

RGB-D Spatial Grounding and Risk-Aware Robot Perception [GitHub](#)

- Implemented a real-time RGB-D pipeline for instance segmentation, target locking, 3D localization, point-cloud processing, object tracking, and nearest-distance risk estimation.
- Organized configurable scripts for Orbbec RGB-D capture, YOLO/SAM2-based segmentation, depth back-projection, mask alignment, offline sequence processing, and visualization reports with JSON metadata.
- The repository separates perception, risk estimation, calibration, utilities, control, tests, and configuration files; recorded sequences include RGB frames, depth frames, masks, camera intrinsics, and per-frame target-state records.

Keywords: RGB-D; point clouds; robot perception; risk estimation

EmbodiedSceneAgent: Structured Scene Memory and Replanning for Embodied Tasks [GitHub](#)

- Built a closed-loop embodied cognition codebase around observation, scene memory, planning, execution, verification, and replanning under changing scene state.
- Implemented structured planner contracts, parser/repair/validation stages, semantic acceptance filters, repeated-no-effect guards, and reproducible smoke and minimal-evaluation scripts.
- Public results report a 73-case diagnostic comparison with improvements in tool-use accuracy, target-match rate, and strict task proxy; the repository explicitly separates proxy metrics from official environment success rates.
- Public documentation includes project overview, showcase cases, limitations, architecture diagrams, setup scripts, and minimal verification commands for reproducing the smoke-test path.
- The README reports accepted revised-plan comparisons on a small controlled set while explicitly treating those cases as diagnostic rather than benchmark-scale generalization evidence.

Keywords: embodied agents; scene memory; verification; replanning

OpenGUI-RL: Verifiable GUI Grounding for Computer-Use Agents [GitHub](#)

- Formulated instruction-conditioned GUI grounding as a scoped one-step contextual-bandit decision from screenshot and instruction to target element, click point, bounding box, and action type.
- Implemented candidate-aware supervised grounding, deterministic reward scoring, reward-labeled candidate generation, reranking analysis, and held-out transfer evaluation across Mind2Web, ScreenSpot-v2, and VisualWebBench.
- The public repository frames results as scoped benchmark evidence for GUI grounding, not as a claim about full browser automation or long-horizon agents.
- Public artifacts include quantitative summaries, tables, figures, data-format examples, reproduction notes, and explicit limitations on dataset redistribution, candidate protocols, and single-step evaluation scope.
- The public result summary highlights OCR/DOM candidate semantics, conditional reranking headroom, point-native transfer, and protocol sensitivity across GUI grounding benchmarks.

Keywords: GUI grounding; verifiable reward; computer-use agents; reranking

- Developed an auditable post-training pipeline for verifiable code generation, centered on MBPP supervised fine-tuning, clean-view data construction, execution- based evaluation, and HumanEval external checking.
- Preserved frozen configs, run-stamped artifacts, data-card notes, non- fine-tuning baselines, training-cost audits, peak GPU memory audits, and a bounded GRPO exploration that is clearly separated from the main SFT-centered conclusion.
- Public archived runs include MBPP and HumanEval evaluation artifacts, clean-view SFT result rows, baseline comparisons, data preprocessing scripts, sandboxed execution checks, and assignment-facing frozen configuration files.
- The repository documents data licensing, processed split sizes, parser/evaluator risks, catastrophic-forgetting sanity checks, training-cost audits, and peak-memory profiling for the promoted runs.

Keywords: LLM post-training; code generation; SFT; execution-based evaluation

ONGOING RESEARCH

- **Temporal interaction understanding in visually complex scenes:** studying how visual and motion cues can support reliable interpretation of fine-grained interactions under occlusion, ambiguity, and changing scene state.
- **Geometry-aware multimodal grounding for embodied perception:** exploring how task context, visual evidence, spatial structure, and uncertainty can support interpretable intermediate representations for reliable decision-making.

Technical details are limited until public manuscripts or preprints are available.

EXPERIENCE

Alibaba.com / Alibaba International Station 2026.05–2026.09
AI Application Development Engineer Intern

ZTO Express Co., Ltd. 2023.06–2023.09
Project-Based Intern

- Worked on operational data analysis for cross-belt sorting equipment, focusing on delayed fault recognition, load fluctuation, resource dispatching, and peak-period scheduling support.
- Used NumPy, Pandas, K-Means, ARIMA, and Hadoop-based workflows for data cleaning, anomaly detection, fault-pattern recognition, load forecasting, and peak-flow analysis; proposed a dynamic allocation strategy for tidal sorting chutes.
- Cleaned 10k+ equipment-operation records with an approximately 89% cleaning rate; the project report recorded a 23% sorting-efficiency improvement and about 89% peak-load forecasting accuracy.
- The resulting solution was later deployed at ZTO’s Chongqing sorting center to support peak-period equipment-load forecasting and scheduling optimization.

TEACHING

Teaching Assistant AY2025–2026, Term 2
CSC1006: Artificial Intelligence for Science and Engineering
The Chinese University of Hong Kong, Shenzhen.

TECHNICAL SKILLS

- **Multimodal and agent systems:** VLMs, GUI agents, OCR/DOM grounding, structured spatial reasoning, grounded decision interfaces.
- **Embodied perception:** RGB-D perception, 3D scene memory, point-cloud processing, long-horizon planning, verification, replanning, failure analysis.
- **Training and evaluation:** PyTorch, PyTorch3D, OpenCV, Open3D, Transformers, PEFT/LoRA, instruction fine-tuning, reward-based reranking, reproducible evaluation pipelines.
- **Programming and data tooling:** Python-based research engineering, NumPy, Pandas, structured configuration, command-line experiment scripts, repository documentation, and LaTeX-based technical writing.
- **Vision and geometry workflows:** segmentation output integration, depth back-projection, point-cloud filtering, object-centric state tracking, spatial-distance computation, and qualitative visualization for debugging.
- **Experiment management:** smoke tests, ablation settings, saved run artifacts, error summaries, report tables, and reproducibility notes for public research-engineering repositories.

AWARDS AND HONORS

- Second Prize, China Undergraduate Mathematical Contest in Modeling, 2023.
- Silver Award, “Internet+” Innovation and Entrepreneurship Competition (Liaoning Division), 2023.
- Outstanding Graduate of Dalian City, 2025.
- Outstanding Student Scholarship and Comprehensive Excellence Scholarship, Dalian Maritime University, 2023 and 2024.
- Outstanding Student Leader / Merit Student / Outstanding League Member, Dalian Maritime University.

LAST UPDATED

May 2026.